



CARE Knowledge Base

Knowledge Engine for CARE, by CARE!

Search by Category

All categories 

Search

Enter a search term, select a category and click **Search** or select an icon below.

- All Articles
- G2
- G3
- Car Park RMS
- RevPlan
- SmartSpace
- Casper
- Optix
- Assets

Dynamic Optimization

Dynamic Optimization :

Dynamic Optimization ensures that RMS controls remain optimal and responsive to rapidly changing business conditions by triggering optimization at key moments during the day, when decisions are most likely to change, and only for the occupancy dates most impacted by the latest booking activity.

User pain points:

With scheduled daytime optimizations (Intraday Processings – IDPs), there is a risk that the RMS may not respond quickly enough to rapidly changing business conditions.

Scheduled daytime optimizations are run for a shorter period than the full optimization window of nightly processing (Business Day End Processing – BDEs), resulting in a risk that the system may not respond quickly enough to changing business conditions further out in the booking window during the daytime.

Event like announcement of a concert or a sporting event at a nearby location can have an immediate impact on the pricing decisions to be made for the days near the events.

Impact:

Categories

Select Category

Recent Posts

[HLTN/hiltonEnableSmallGroupProductsAndInventoryLimitForNewPropertyStep](#)

[Service Recovery Procedure – IDeaSOperaAgent Service in “Stopping” State](#)

[Processing Failures Due to Data Type Mismatch for MGM Resorts](#)

[Dynamic Optimization](#)

[HLTN/Handling Error for SAS Dataset Encoding Job](#)

This solution has an impact on the RMS' optimization trigger and optimization window size; however, it does not change the optimization process itself. The solution does not change the inputs nor the outputs of the process either.

Benefits

- Ensures RMS controls remain optimal at all times.
- Improves RMS responsiveness to real-time booking activity and changing business conditions.
- Enables the system to react in real time to changing booking activity further out in the booking window.
- The ondemand IDP can only optimize for the IDP window which may not cover the dates of a new pricing affecting event like a concert or a sporting event

Inputs:

- Incremental solds (Real time bookings) – Can be Transient/ Group
- Reference price
- Delta LRV

Parameters:

pacman.feature.populateReferencePriceLatest	BDE, CDP jobs	reference_price_latest
pacman.feature.sendSnsNotificationOnConfigChange	Datadog, G3	
pacman.PreProduction.operationTypeAgnosticUploadEnabled	Decision delivery jobs	
NGI integration – dynamicOptimization	NGI, G3	
pacman.feature.dynamicOptimizationEnabled	Functionality	
pacman. <u>optimization</u> .nextIdpWindowOverride	Max Window date	
pacman.feature.saveDecisionVolume	Count of decision post BDE/IDP	Decision_volume

Process

- Basic check to eliminate invalid properties – Check last BDE/IDP time, already calibrated etc.
- Run evaluation
- get incremental solds, reference rate, lrv delta values and calibrated potential from dynamoDB
- incremental sold size
- Reference rate size → 365 occ Date * number of room class = 7300
- LRV values → 365 * number of roomclass
- Trigger IDP if needed – Send SNS notification which would be subscribed by G3

Recent Comments

Pranav Thakar on [Hilton/Processing Failure – non positive offset value for Product](#)

Ranjit on [Hilton/Processing Failure – non positive offset value for Product](#)

Ranjit on [Service Recovery Procedure – IDeaSOperaAgent Service in “Stopping” State](#)

Ranjit on [Processing Failures Due to Data Type Mismatch for MGM Resorts](#)

Ranjit on [Accor Client Handling Process Guide](#)

Outputs:

Whether IDP should trigger – Yes / NO

Max occupancy date value

Rollback:

In case of issues, please raise this via Teams channel / SFDC's.

Point of Contact (Technology team) : Team Heisenberg (Sendhil, Rohan, Mehal, Atul)

Product team : Maureen, Andres

FAQ's

How does the “Potential Variable” work?

The “Potential” variable combines key adjustments like booking activity, room value (LRV) and fair market value into one easy-to-read number for each room class and date range. This helps the system quickly see whether the current activity is above the normal levels and decisions would likely change by triggering optimization.

Where will the solution's configurations and parameters sit? And who would have access to edit/modify?

Configuration and parameters are planned to be set and calibrated only by IDEaS Analytics via the system's backend.

Is Dynamic Optimization triggered by Rate Shopping data Updates?

Booking activity is the only data that will initially trigger optimization. Rate shop or other data inputs (such as Demand360) will not trigger optimization, nor will new data be fetched based on an optimization being triggered. Optimization will consume the latest data available as it does today.

Does Dynamic Optimization replace IDPs or BDEs?

At this time, it is planned for Dynamic Optimizations to complement IDPs and BDEs, not replace them.

How often will the solution check if optimization needs to be triggered?

The solution is expected to analyze booking activity every 10 minutes to determine if an optimization is needed. 10 minute frequency was defined due to experience with other integration-related processes, anything more frequent is not expected to provide major activity updates that would justify running optimization.

What is the maximum number of optimizations per day?

The maximum number of optimizations per day that will be supported is 8.

Is there a minimum time gap between optimizations? If so, what is it?

60 minutes between system optimizations.

Will there be a change in system lock time during dynamic optimization processing?

There will be a system lock during dynamic optimization, similar to IDP.

Why does Dynamic Optimization always start at DTA 0 even if the trigger booking activity might be further in the future?

Considering the RMS's optimization network effect, it is important to start from DTA0 to minimize the risk of demand being truncated. Additionally, this approach offers the advantage of updating occupancy dates closer to the arrival date, incorporating any updates from Rate Shopping or other external forecast data.

PPT :

Dynamic Optimization

👍 0

 Vinayak Salunkhe

 08/08/2025

 G3 Processes, Hilton Monitoring Documents

 No Comments

← HLTN/Handling Error for SAS Dataset Encoding Job

Processing Failures Due to Data Type Mismatch for MGM Resorts →

Leave a Reply

Your email address will not be published. Required fields are marked *

Comment *

Name *

Email *

Website

☐ Save my name, email, and website in this browser for the next time I comment.

Post Comment